An Empirical Analysis of Transfer for Dynamic Path Planning

Anonymous submission

Abstract

We study the dynamic path planning problem, where an agent needs to plan a route in the presence of moving obstacles. In particular, we study whether heuristics learned in training maps can be transferred—zero-shot or from a few labeled target maps—to new environments for this task. We show that, within budgeted time-expanded A* and with the right training objective and planner integration, a frozen heuristic transfers zero-shot across six procedural continuous dynamic suites. Even given no information about future obstacle motion, it improves success over a geometric anchor while substantially reducing search effort. Selected discrete-grid experiments show a smaller rank-only effect. In K-indexed adaptation, low-rank updates better preserve success at one labeled target map, while full fine-tuning is more search-efficient by sixteen maps at near-parity success. A separate matched-label factorial shows that distributing a fixed label count across more distinct target worlds prevents the held-out success degradation that concentrated supervision causes. Further, across comparable backbones no architecture is uniformly best: Multilayer Perceptrons, recurrent models, Hierarchical Reasoning Models and U-Net architectures all produce useful learned heuristics in different tested formulations, while hierarchy-specific gains do not persist.

Introduction

Dynamic path planning searches over both configuration and time: when obstacles move, a planner must consider waiting, detouring, and timing, and the search space grows accordingly (Silver 2005; Phillips and Likhachev 2011). The geometric admissible heuristics that make A* (Hart, Nilsson, and Raphael 1968) efficient on static maps ignore exactly these temporal effects, so search effort grows sharply with moving obstacles. Learned heuristics can reduce search effort (Arfaee, Zilles, and Holte 2011; Bhardwaj, Choudhury, and Scherer 2017; Yonetani et al. 2021), but a deployed planner benefits only if a heuristic trained on one set of maps remains useful on maps it has never seen. We study zero-shot and few-shot transfer of learned heuristics to unseen environments for dynamic path planning; budgeted time-expanded A* serves throughout as a controlled measurement substrate, not a proposed replacement for dedicated dynamic planners.

We organize the experiments around three factors—the training formulation, the way the prediction enters the planner, and the distribution of target supervision—each evaluated first on static suites, where supervision is exact, then under deterministic dynamics. Comparisons use matched controls throughout—input-matched twins, identical weights under different integrations, anchor-calibrated budgets, map-level paired inference (Henderson et al. 2018; Agarwal et al. 2021; Pineau et al. 2021)—with confirmatory comparisons, units, and criteria specified before test evaluation. Across the tested settings, representation, planner integration, and supervision regime explain the observed transfer outcomes more consistently than model family.

Our contributions are:

- **Dynamic zero-shot transfer with one fixed learned heuristic.** A single U-Net heuristic improves success on all six dynamic suites of a 50-map-per-suite confirmation cohort ($+0.18$ to $+0.84$; every paired map-level CI excludes zero) and cuts matched search effort by 73–95%. The model was development-selected, frozen before the confirmation cohort was generated, and given no future-motion input. Three suites are held out: one distinct topology and two parameter/scale shifts; the remaining three are disjoint maps from trained families.
- **Planner integration changes whether transferred predictions reduce search.** Changing the input–target–output formulation rescues a collapsed recurrent model; changing only the planner integration turns an inert discrete residual into useful guidance; and sharing one search state turns an exact oracle from a six-map loss into a six-map win. A negative planning result therefore does not isolate the model unless target and search interface are held fixed.
- **Distinct-world coverage, not label count, governs held-out success in the tested matched-label factorial.** With scarce static supervision, low-rank adaptation preserves the transferred prior while full fine-tuning is unstable; with more supervision, full fine-tuning often overtakes it on search effort at near-parity success (direct map-paired contrasts at both ends). A matched-label factorial ties the low-supervision failure to concentration in too few

distinct worlds, not raw label count. Distributing the same labels across more worlds rescues full fine-tuning, and targeted world-draw replications preserve the coverage contrasts. Low-rank adaptation shows the least frequent and smallest degradation across the tested cells, though one redraw produces a significant LoRA degradation.

Additional controls define the scope. Under the tested weighted-A* grid, the learned heuristic keeps a success advantage only on spiral and lower matched expansions on four suites, at higher path cost on two. Unbudgeted safe-interval planning solves every feasible-labeled instance optimally under the implemented collision model, and the zero-shot success advantage does not reproduce in a small MovingAI-derived boundary test (analyzed post hoc below). Boundary experiments: future-window inputs, adapter interpolation, and hierarchy-specific depth add no consistent value; a restricted-observation heuristic with one Bellman backup and frozen scale calibration beats a complete-map comparator on expansions.

Problem Formulation and Learned-Heuristic Interfaces

Planning problems. In the continuous setting, a point robot plans on a probabilistic roadmap (PRM; Kavraki et al. 1996) built over a square world of side length $L{=}1.0$ (2.0 for the large-room family); units are abstract. Each roadmap has 192 nodes in total—the start and goal at indices 0 and 1 plus 190 sampled free-space nodes—connected by $k{=}7$ nearest-neighbor candidate edges per node (doubled for the start and goal, then symmetrized into an undirected graph). Edges are validated by sampled collision checks at resolution $0.01L$, and unconnected worlds are rejected. Static search states are roadmap nodes, edge costs are Euclidean lengths $\ell(e)$, and the anchor heuristic is Euclidean distance to the goal, $h_0(v) = \|x_v - x_g\|$. In the dynamic setting, obstacles move on deterministic trajectories and the search state is a pair (v, t) of node and discrete time step of duration Δt. The planner may traverse an edge, taking $\tau(e) = \lceil \ell(e)/(v_{\text{agent}}\Delta t) \rceil$ steps, or wait one step; transitions are validated against the obstacle trajectories by sampled collision checks along the edge and interval. Cost is arrival time in steps, search runs to a horizon $t_{\max}$, and the anchor is the travel-time lower bound $h_0(v,t) = \|x_v - x_g\|/(v_{\text{agent}}\Delta t)$, constant in t, so anchor and cost share units. In the discrete setting, agents plan on occupancy grids of $20{\times}20$ to $256{\times}256$ cells with Manhattan distance as the anchor.

Supervision targets. The exact cost-to-go h^* comes from graph Dijkstra on the static roadmap and from backward space–time Dijkstra over (v, t) dynamically. The regression target is the normalized anchor residual $\bar{\delta}^*(s) = \text{clip}(h^*(s) - h_0(s),\, 0,\, cT)\,/T$. Here T is the normalization scale ($T{=}L$ statically; dynamically $T = L/(v_{\text{agent}}\Delta t)$, the map-crossing time in steps, identical across suites by construction), the cap $c{=}4.0$ applies after normalization, and unreachable states are masked from the loss. Scalar models predict $\bar{\delta}_\theta$ from per-node feature tokens; field models predict a goal-conditioned residual raster (eight input channels; appendix) sampled at node positions. At integration the prediction is rescaled: $\hat{h}_\theta(s) = h_0(s) + T \cdot \text{clip}(\bar{\delta}_\theta(s), 0, c)$.

Planner integrations. Under *additive* integration the planner orders states by $f(s) = g(s) + \hat{h}_\theta(s)$, sacrificing admissibility for guidance; empirical path-cost ratios are reported wherever additive integration is used. The classical control is weighted A* (Pohl 1970), which orders states by $f(s) = g(s) + w_h\, h_0(s)$ with a per-suite success-tuned inflation w_h. Under *focal (rank-only)* integration (Pohl 1970; Pearl and Kim 1982), the anchor priority $f_0(s) = g(s) + h_0(s)$ defines the band $\text{FOCAL}_w = \{s \in \text{OPEN} : f_0(s) \le w \cdot \min_{u \in \text{OPEN}} f_0(u)\}$, and the learned prediction orders expansions only within the band. At $w{=}1.0$ the learned score merely breaks ties among minimum-priority states, preserving the anchor's path-cost optimality when search completes (finite budgets can still change success). In the restricted-observation study, learned predictions serve as exit values of a radius-bounded local subgraph; one inference-time Bellman backup over that subgraph produces the heuristic, with a scale calibrated once on a disjoint block.

Transfer protocol. Zero-shot transfer applies a trained-and-fixed heuristic to unseen maps and unseen map families with no target supervision. Few-shot adaptation draws K labeled target maps and compares low-rank adaptation (LoRA; Hu et al. 2022) (rank 8 for the HRM, rank 24 for the wider ON-LSTM), full fine-tuning, and training from scratch. LoRA and full fine-tuning start from the same trained checkpoint; scratch uses the same architecture and schedule from random initialization.

Experimental Setup

Environment families. A *family* is a procedural map generator; a *suite* is a named evaluation condition; a *world* is one sampled environment. Continuous worlds come from our procedural generator (specification tables in the appendix; code in the anonymized artifact archive). The static hard families are wall constructions—*maze/dense maze* (corridor mazes; the dense variant narrows gaps to $0.115L$), *rooms/large rooms* (partition walls with doorways; the large variant doubles the side length), *spiral*, and *bugtrap* (generator summaries in the appendix; exact implementation in the artifact archive). Static training uses the maze, rooms, and spiral families; dense maze, bugtrap, and large rooms are held out (bugtrap: distinct topology; dense maze/large rooms: gap- and scale-shifted variants of trained generators). Dynamic suites add 3–5 moving circular obstacles per world that sweep line segments with triangle-wave motion (radius 0.055–0.080L, span 0.36–0.50L, period 0.14–0.30 of the horizon; per-suite parameters in the appendix), plus a *crossing* open-arena suite as a timing control. *Dynamic training draws worlds only from the dynamic maze, rooms, and spiral families; dynamic dense maze, crossing, and large rooms are held out*—crossing a distinct open-arena topology, dense maze narrower corridors and adjusted patrollers, large rooms doubled scale with a faster agent. Three of the six test suites therefore share a generator with training (disjoint maps); three are suite-level shifts. No learned model observes a test world during training; evaluation cohorts come from disjoint seed ranges, checked by world fingerprints.

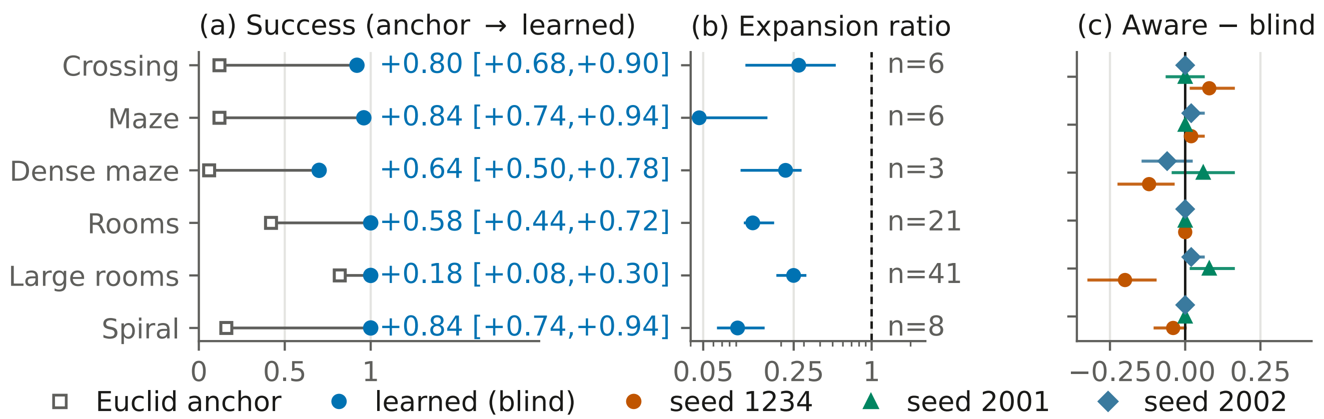

Figure 1: Dynamic zero-shot transfer of the fixed blind U-Net (no future-motion input) on the 50-map-per-suite confirmation cohort (suite taxonomy in the text). (a) Anchor and learned success per suite; labels: paired mean delta with map-level 95% CI (all six exclude zero; marginal). (b) Matched-solved expansion ratios, conditional on joint success (log axis, map-level 95% CIs); n = jointly solved count (dense maze n=3 descriptive). (c) Aware−blind success differences for the three training pipelines; the large-rooms cell flips sign.

Models. Scalar models consume per-node token sequences (62 tokens of 16 dimensions; appendix) and include an MLP, an LSTM (Hochreiter and Schmidhuber 1997), an ON-LSTM (Shen et al. 2019) (1,252,481 parameters), and a Hierarchical Reasoning Model (HRM; Wang et al. 2025) (2,158,529 parameters) with an adaptive-computation port (Graves 2016; Banino, Balaguer, and Blundell 2021). Field models consume the eight-channel 64×64 raster and include a FiLM-conditioned U-Net (Ronneberger, Fischer, and Brox 2015; Perez et al. 2018) (3,702,370 parameters), a field HRM (4,016,642), and a field ON-LSTM (1,448,578); dynamic variants append W future-occupancy channels (W=8 aware = ground-truth occupancy at offsets t+1, . . . , t+8; W=0 blind). Unless a specific experiment states otherwise, the continuous residual models train with AdamW (learning rate 2×10^{-4}, weight decay 10^{-4}, batch 256, gradient clipping 1.0) on a smooth-L1 loss over the capped normalized residual; primary architecture and optimizer tables are in the appendix; complete constructors ship in the artifact. Within each direct architecture comparison, models are trained under one matched recipe on identical data.

Evaluation. Each suite is evaluated at a per-suite expansion budget selected by anchor-only calibration: the grid budget whose anchor success is closest to pre-specified targets (evenly spaced in $[0.45, 0.70]$; ties smaller) is fixed before any learned method is compared. The smallest selected budget is binding. The grid is coarse, so realized anchor success can fall outside the target band (0.05–0.75 dynamic at calibration, 0.25–0.62 static; grid and realized rates in the appendix). Static calibration shares worlds with evaluation (anchor-only, never touching learned outcomes; disclosed); we therefore call those runs calibrated benchmark evaluations. The dynamic 50-map cohort and the 144-map restricted-observation cohort are *confirmation* cohorts: their budgets were frozen before the cohorts were generated. Success means reaching the goal within the budget; appendix curves up to $4\times$ binding mitigate single-threshold dependence. Search effort is the *matched-solved expansion ratio*: the median over jointly solved maps of the per-map ratio of learned to anchor expansions. Because it conditions on joint success, it can favor methods that solve only easier maps; success is therefore reported first, with path-cost ratios reported separately (appendix).

Statistics. Success comparisons use paired McNemar tests with Benjamini–Hochberg correction within each experiment's suite family (adjusted values reported as q). Interval estimates are percentile bootstraps with 10,000 resamples; the resampling unit is the evaluation map, and repeated adaptation or model seeds are averaged within maps before resampling; quoted intervals reflect map uncertainty conditional on the evaluated seeds. Per-suite bootstrap intervals (as in Figure 1) are marginal; the corrected McNemar tests are primary (full policy in the appendix). Method-versus-method claims use direct map-paired contrasts. Unless noted, continuous experiments use one primary training seed; the dynamic result adds two retrained pipelines.

Dynamic Zero-Shot Transfer

Setting and fixed-model protocol. The dynamic experiment trains on the dynamic maze, rooms, and spiral families and evaluates on all six dynamic suites—three sharing a training generator (disjoint maps) and three held out as above (dense maze, crossing, large rooms). The training run keeps 53 usable worlds of 72 candidates, 24 per family (usable: builds, roadmap connects, space–time solvable at t=0); their tensors hold 1,129,536 node–time slots, 821,850 (72.8%) of them supervised (appendix). The original 20-map-per-suite cohort served as *development* data: on it we selected the field U-Net *blind* twin and froze its checkpoint, additive integration, and per-suite budgets. The 50-map-per-suite *confirmation* cohort was then generated from a pre-specified seed range, map-disjoint by fingerprint check, with no further choices. Selection still happened, on development data;

the protocol protects only the confirmation cohort. The blind heuristic receives only current-time occupancy channels; the space–time planner and collision checker always use the true deterministic trajectories when validating (v, t) transitions, so the aware/blind ablation changes only the heuristic's input.

Confirmation result. On the confirmation cohort, the fixed blind U-Net raises success on all six suites: deltas $+0.18$ to $+0.84$, every paired map-level 95% CI excluding zero (exact McNemar $q \leq 0.0039$; every discordant map favors the learned heuristic). Matched expansion-ratio medians are 0.047–0.273, a 73–95% effort reduction on jointly solved maps (Figure 1; joint counts 3–41 per suite). Empirical path-cost ratios on solved maps stay within per-suite means of 1.008–1.164, and the pattern reproduces the disjoint development cohort's (appendix). Gains on the three held-out suites ($+0.64$/$+0.80$/$+0.18$) establish transfer under topology and parameter/scale shift; the rest, map-level generalization within trained families.

Independent retraining robustness. Two additional full training pipelines with independent seeds reproduce the success pattern on the same held-out cohort; we disclose that they used a larger world-collection budget (139/149 usable worlds versus 53; appendix), so seed and data scale vary together. Across the three pipelines, 17 of 18 pipeline–suite paired CIs exclude zero (deltas $+0.10$ to $+0.88$; the exception is seed 2001's near-ceiling large-rooms cell, anchor 0.82). Effort magnitudes are stable across pipelines on three suites (medians 0.047–0.135) and pipeline-variable on the other three (0.216–0.625), quoted as across-pipeline ranges there; the retrained models are evaluated on the same 50 maps (retraining robustness, not cohort replication).

Tuned weighted A* control. Because the anchor is deliberately simple, we also compare against weighted A* ($w_h h_0$), success-tuned per suite over $w_h \in \{1.1, 1.2, 1.5, 2, 3, 5\}$ on the development cohort (rule frozen in advance: highest success, ties smaller) and evaluated once on the confirmation cohort. The control was specified after the learned results were known; tuning touched only development data (appendix). The comparison yields a three-way tradeoff rather than dominance (Table 1). *Success:* no significant difference is detected on five suites (at most one discordant map on four; crossing's four-map weighted advantage has q=0.375 under the exact McNemar policy). The learned heuristic holds the one BH-significant advantage, on spiral ($+0.30$ $[+0.18, +0.42]$; 15/0 discordant, q=3.7×10^{-4}). *Effort:* learned-to-weighted expansion medians are below parity everywhere, with the marginal 95% interval excluding parity on four of six suites (0.217–0.728); crossing and dense-maze intervals include parity (descriptive—the corrected success tests carry the inferential policy). *Path quality:* on jointly solved maps, the additive learned heuristic pays a premium over tuned inflation on crossing ($+0.156$ $[+0.116, +0.200]$) and large rooms ($+0.080$ $[+0.058, +0.105]$), both paired intervals excluding zero. The other four suites stay within ± 0.005, intervals covering zero. Because three suites selected the grid maximum w_h=5, we also ran a disclosed post-hoc extension to $w_h \in \{7, 10\}$ (design frozen before execution). Five selections are unchanged; in particular spiral's development success stays 0.80, so its learned advantage is robust to the tested extension. Dense maze re-selects w_h=7 (development 19/20 versus 18/20), changing its point estimates but no inferential conclusion: success 0.74 versus 0.70 against both the frozen rows and the learned arm (exact $p \geq 0.5$), effort at parity (median 1.40 $[0.91, 1.67]$), path quality within ± 0.007. Full per-suite intervals are in the appendix.

Unbudgeted SIPP nearly saturates the substrate. We also implemented discrete-time SIPP (Phillips and Likhachev 2011) on the identical substrate with the same collision predicates and the anchor heuristic, gated by a hard correctness check: on all 300 confirmation instances, SIPP's earliest arrival exactly equals the space–time optimum. Under the implemented sampled-collision model, unbudgeted SIPP solves every instance labeled feasible at the backward-Dijkstra optimal arrival. Its per-suite success, 0.94–1.00, equals the substrate's feasibility ceiling exactly, with small expansion counts in its own interval-state unit (medians 80–290; never merged with (v, t) expansions). Wall times: SIPP 0.9–1.8 s per map (full-cohort means); learned 1.18 s end-to-end (1.02 s table + 0.15 s search) and weighted A* 0.23 s on the timing sample (different protocols; appendix). This confines the learned result to budgeted time-expanded search: it restructures effort there; it does not outperform a dedicated dynamic planner on this deterministic substrate (appendix table).

External-geometry boundary. To test the claim beyond our own generators, six MovingAI maps (Sturtevant 2012) (three street, three dungeon) were converted to the canonical substrate: 64×64 occupancy, rectangle-decomposed obstacles, and the trained maze-family patroller dynamics. The frozen heuristic was then evaluated under the frozen calibration/tuning protocol. One dungeon map yields no usable instances, so five maps contribute (appendix). The success advantage does not transfer. The learned heuristic never significantly beats the anchor (deltas -0.08 on both groups), and on dungeon instances it loses to tuned weighted A* (-0.28, 0/7 discordant, exact p=0.016; instance-level, unadjusted, nested in two maps) while expanding more nodes (matched ratio 2.20 $[1.25, 6.36]$). It pays path-quality premiums with intervals excluding zero on both groups; only the street-map matched-effort advantage over the anchor survives (0.48 $[0.28, 0.64]$). The zero-shot success claim is therefore bounded to the procedural families and their tested shifts. A post-hoc residual analysis (appendix) associates the failure with miscalibration: predicted-vs-true residual correlation falls from 0.73–0.98 in-distribution to 0.24–0.39 on the converted maps, with the over-prediction bias the inflate-only interface punishes. Exploratory few-shot adaptation on 2–8 development instances from the same source-map groups restores dungeon success to 0.90–1.00 and street to classical parity, while single-instance adaptation degrades street. The pattern is qualitatively consistent with the factorial's narrow-coverage degradation, though not a matched replication.

Future-window inputs provide no consistent benefit. The primary model is already the blind twin. The aware twin adds eight ground-truth future-occupancy input channels (3,704,674 vs. 3,702,370 parameters, first layer only); architecture and training are otherwise matched. On the held-out cohort the contrast is suite-dependent in both directions

Suite	w_h	Success learned	Success WA*	Expansions learned/WA* [95% CI]	Cost learned	Cost WA*
Crossing	1.5	0.92	1.00	0.825 [0.531, 1.167]	1.164	1.008
Maze	5	0.96	0.96	0.601 [0.440, 0.835]	1.013	1.017
Dense maze	5	0.70	0.70	0.982 [0.823, 1.089]	1.008	1.006
Rooms	3	1.00	0.98	0.554 [0.439, 0.798]	1.019	1.015
Large rooms	1.5	1.00	0.98	0.728 [0.536, 0.966]	1.083	1.003
Spiral	5	**1.00**	0.70	0.217 [0.183, 0.312]	1.014	1.013

Table 1: Fixed learned heuristic versus per-suite success-tuned weighted A* (original grid), confirmation cohort at binding budgets. Expansions: median learned-to-weighted ratio on jointly solved maps [map-bootstrap 95% CI]. Cost: mean arrival over optimal arrival on the same jointly solved maps (paired intervals in text). Bold: the only BH-significant success difference (exact McNemar q=3.7×10^{-4}; crossing's four-map gap q=0.375). A disclosed post-hoc grid extension re-selects w_h=7 on dense maze but changes no inferential conclusion (success 0.74 vs. 0.70, $p \geq 0.5$; appendix).

and unstable across pipelines. Of 18 contrasts, the paired interval excludes zero for the aware twin in two, the blind twin in two, and neither in fourteen; the excluded cells fall in different suites for different pipelines (large rooms flips sign). The canonical pipeline also trained on fewer worlds, but the two collection-matched retrains still disagree (Figure 1(c); appendix). The original 24 suite-by-backbone cells point the same way, and after full fine-tuning at K=16, eight of nine success differences are non-positive (the ninth includes zero). We report a failure to observe a systematic future-window benefit under deterministic dynamics—not equivalence, and not harm; stochastic motion is untested.

Effects of Supervision Target and Planner Integration

Static family transfer. On the six calibrated static suites, the transferred field HRM solves 0.75–1.00 of test maps against anchor success of 0.25–0.58 (BH-adjusted q from below 0.001 to 0.025 on four suites; matched expansion ratios 0.521–0.850). Three of six suites are held out as above, and pooled source-family models improve success on every held-out family before any target labels.

Supervised target. On the calibrated hard static maps, a pooled ON-LSTM scalar residual model raises hard-maze success from 0.525 to 1.000 over the anchor ($q = 4.3 \times 10^{-11}$). The parameter-matched scalar HRM instead collapses to a constant prediction at the residual cap and merely matches the baseline (lower learning rates and a soft cap reproduce the collapse). Retaining the HRM while changing input representation, supervised target, and output head to the raster residual field raises hard-maze success from 0.625 to 0.975 ($q = 3.4 \times 10^{-4}$); a later experiment shows a well-trained scalar HRM is also viable (ratios 0.427–0.822). This localizes the failure to the scalar training formulation, though the field intervention changes several coupled components.

Planner integration. The preferred integration differs between the two tested substrates. On continuous roadmaps, additive residuals exploit the gap the Euclidean anchor leaves (the field-HRM ratios above, at empirical path-cost ratios of roughly 1.02–1.14), while rank-only variants stay closer to plain anchor search. On discrete grids the learned signal is demonstrably informative: the pooled residual correlates 0.987–0.994 with the true residual across map sizes. But its magnitude is scale-flat (predicted/true mean 0.73 to 0.19 as maps grow), and every additive configuration in the 22-suite run is at or below its matched baseline. We then reuse the *same* trained predictions only to rank expansions inside a w=1.0 focal band: expansions fall 6–15% with no observed success regression (Chrestien et al. 2023)—changing only the integration recovered the ordering signal. We interpret the cross-substrate pattern—additive helping against a looser anchor, ranking against a tighter one—as a supported hypothesis; anchor tightness itself is not directly measured.

Oracle integration control. An oracle control removes model-estimation error. The same exact value ranker loses all six development-map comparisons under a fresh certifier search that discards the incumbent's work (141.2 vs. 129.7 expansions), and wins all six when sharing one search queue and g-state (122.8). Identical values, only search-state reuse changed (a development mechanism test).

Few-Shot Adaptation

Direct contrasts: LoRA protects at K=1; full fine-tuning overtakes on effort by K=16. At K=1 the map-paired full-FT−LoRA success difference is negative in all six target/backbone cells, with the 95% interval excluding zero in four (worst -0.600 $[-0.773, -0.407]$ on large rooms/HRM). LoRA retains the zero-shot gain (dense maze ratio 0.686, success $+0.467$); full fine-tuning reaches ratio 1.293 on large rooms, and scratch is harmful on dense maze (-0.167, interval excluding zero). At K=16 the contrast reverses on effort: the paired expansion-ratio difference favors full fine-tuning in all six cells, with the map-bootstrap interval excluding zero in five (up to -0.366 $[-0.489, -0.235]$). Success is at near-parity (one cell's interval favors full fine-tuning). A 27-cell matched-compute ablation finds bounded and unbounded LoRA nearly identical (median map-level ratio delta 0.000, maximum 0.037), so the output clamp is not the primary cause of the LoRA plateau. LoRA is not universally protective either (bugtrap ratio 1.153 by K=32). Dynamic maps are label-dense ($\sim$15,507 supervised states per world), and at dynamic K=1 full fine-tuning is not systematically catastrophic while LoRA continues to improve rather than merely preserve the base. Full per-cell contrasts and illustrative K-indexed curves are in the appendix.

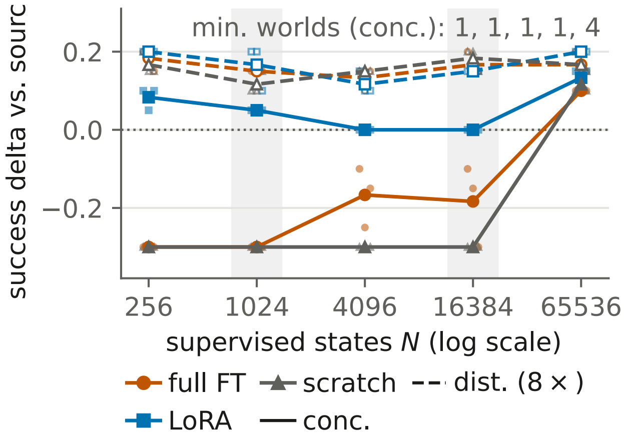

Figure 2: The matched-label factorial, dynamic substrate (static panel and tables in the appendix): success delta vs. the frozen zero-shot source at fixed state count N, frozen test cohort, binding budgets. Solid: concentrated (fewest worlds supplying N; counts annotated); dashed: the same N over $8\times$ as many worlds; small markers: the three adaptation seeds (one sampled world set per cell). Concentrated supervision drops held-out success for full fine-tuning and scratch; distributing the same labels prevents it; recovery at N=65,536 coincides with the world minimum rising to 4. Shaded bands mark N levels with targeted independent world redraws, which preserve positive coverage contrasts there; one redraw LoRA cell degrades (appendix).

A matched-label factorial separates label count from distinct-world coverage. The K-indexed protocol confounds two quantities: adding maps adds labels and worlds together. A preregistered factorial therefore holds the number of supervised states N fixed (256 to 65,536 by factors of 4) and varies only whether those states come from the fewest worlds that can supply them (*concentrated*) or from eight times as many (*distributed*). Both domains run under matched optimizer steps (Figure 2; protocol and tables in the appendix). Each cell adapts one deterministically sampled world set (seeds vary initialization and batch order), with targeted redraws reported separately (appendix). The preregistered primary hypothesis—a full-fine-tuning$\times \log N$ effort crossover—is rejected (interaction -0.0025 $[-0.0087, +0.0028]$); the coverage readout below is the prespecified secondary endpoint. Concentrated supervision causes a held-out success drop for full fine-tuning and scratch in both domains: -0.300 in every seed at dynamic concentrated $N \leq 1{,}024$, and still negative in every seed at N=16,384 from a single world (one of three seeds excludes zero). Distributing the *same* states across more worlds prevents the drop ($+0.15$ to $+0.20$; Figure 2). Direct distributed−concentrated contrasts for all 30 cells carry BH-corrected sign-flip tests (added post hoc during review response; appendix): $q<0.05$ for full fine-tuning and scratch in all 10 cells where concentration forces at most two distinct worlds, and in none of the 10 where it forces four or more (threshold descriptive). No significant LoRA success degradation appears in the original 60 cells (worst -0.067 $[-0.167, 0.000]$); absent a noninferiority margin, this is a failure to observe degradation, not equivalence. Targeted world-draw replications at the decisive cells make the caveat concrete: all 18 replicated coverage contrasts are positive, excluding zero for scratch in all six and for full fine-tuning in five of six (the second draw's static contrast touches zero). One redraw LoRA cell does degrade, its interval excluding zero (static concentrated N=256: -0.133 $[-0.267, -0.033]$, still the least-degraded method in that cell; appendix). On the static matched expansion ratios (the domain with adequate jointly solved counts), full fine-tuning is at or below LoRA at every tested N, consistent with the rejected interaction above. Dynamic effort cells contain one jointly solved map and support no inference; because ratios condition on solved maps, degraded-success cells can look spuriously efficient. Success is therefore the factorial's primary endpoint. The protection low-rank adaptation provides at low supervision appears in *success* and tracks distinct-world coverage rather than label count in this factorial.

Scope and Boundary Results

Discrete transfer. The pooled HRM base improves mean grid success $0.591 \to 0.612$ with lower mean expansions (descriptive), and adapters do not improve broadly. A 3–8-seed rank-only pilot on selected maps finds 6–15% expansion reductions at w=1.0 with equal or higher observed success (no full-suite confirmation).

Architecture and hierarchical depth. Descriptor-weighted adapter composition routes correctly yet never significantly beats the pooled zero-shot model (appendix) (Wortsman et al. 2022; Ilharco et al. 2023). Architecture rankings differ across tasks and input representations (the U-Net is strongest in the dynamic field study; the HRM holds the lowest single-suite dynamic ratio; the ON-LSTM leads the early calibrated scalar stage). In compositional missions with verified oracle headroom (three-seed grids; appendix), no structured model beats the MLP at two depths with a non-decreasing gap, and learned halting moves *opposite* to the depth hypothesis (map-level $\rho = -0.578$, permutation $p = 5 \times 10^{-5}$). Five of 33 recurrent cells suffer constant-prediction collapse and are excluded from capacity conclusions. Follow-ups on persistent memory, iterative refinement, hierarchical substitution, and persistent planning-time search state are also negative (appendix). These are formulation-specific negatives, not claims that recurrence, memory, or hierarchy are universally ineffective (Jolicoeur-Martineau 2025; Czechowski et al. 2021).

Restricted-observation transfer. Deployed agents rarely observe complete maps, so a final static study restricts the model to a radius-0.20 local subgraph, trains it from behavior-derived returns rather than shortest-path labels, and integrates it through one inference-time Bellman backup. The frozen comparison runs on a disjoint 144-map cohort; its path-quality gate is the comparator-relative criterion adopted after the original absolute ceiling failed in development (appendix). The local MLP reduces pooled expansions against the complete-map field HRM by 12.96 (95% CI $[-16.30, -9.74]$; 109/3/32 W/T/L) at better empirical path quality (mean/maximum cost ratios 1.024/1.162 versus 1.031/1.335). A separate $w{=}1.10$ bounded control passes all 144 certificates. Sequential ablations locate the sign change at the backup (static insertion loses by $+16$; one backup flips the point estimate to -1.21, interval crossing zero; frozen scale calibration then yields the confirmed reduction).

Related Work

Learned guidance for search. Learned heuristics arise from bootstrapping (Arfaee, Zilles, and Holte 2011), oracle imitation (Bhardwaj, Choudhury, and Scherer 2017), differentiable search (Yonetani et al. 2021), grid transformers (Kirilenko et al. 2023), classical-planning objectives (Ferber, Helmert, and Hoffmann 2020), and ranking (Chrestien et al. 2023). We retain classical graph search throughout and jointly evaluate cross-family and dynamic transfer of identical trained models and additive-versus-rank-only integration of the *same* predictions under a shared planner and information set (Pohl 1970; Pearl and Kim 1982; Aine et al. 2014).

Dynamic, real-time, and motion planning. The dynamic substrate uses time-expanded space–time search in the spirit of cooperative A* (Silver 2005); safe-interval planning (Phillips and Likhachev 2011) is a related, faster alternative, which we evaluate as an unbudgeted optimal control. The restricted-observation study relates to real-time heuristic search (Korf 1990; Koenig and Likhachev 2006); its one-step backup resembles a single value-iteration-network update (Tamar et al. 2016; Lee et al. 2018) and neural algorithm execution (Veličković and Blundell 2021), but runs once, at inference, over a radius-bounded subgraph, to integrate a transferred prediction. Learned samplers and end-to-end motion planners (Ichter, Harrison, and Pavone 2018; Qureshi et al. 2021) replace the planner itself; we keep a fixed classical planner and learn only its guidance on a PRM (Kavraki et al. 1996).

Adaptation and recursive architectures. LoRA (Hu et al. 2022), model soups (Wortsman et al. 2022), and task arithmetic (Ilharco et al. 2023) motivate the low-rank and composition studies; the contribution is the low-rank-versus-full-rank comparison as a function of label availability and, through the factorial, supervision distribution across worlds. HRM (Wang et al. 2025), adaptive computation (Graves 2016; Banino, Balaguer, and Blundell 2021), and simplified recursive models (Jolicoeur-Martineau 2025) motivate the architecture experiments; our results concern heuristic regression and integration, not their benchmarks.

Limitations

Design and selection. Most continuous primary results are single-GPU validations with one primary training seed, and several experiments reuse trained bases or test maps rather than replicating independently. Only the 50-map cohort is selection-free; static calibration shares worlds with evaluation (anchor-only; budget curves mitigate); baselines are the anchor, weighted A*, and SIPP—external *learned* planners are not compared. The retrained pipelines used larger collections (139/149 vs. 53 worlds; seed and scale vary together); the grid extension was post hoc. A suite-indexing defect made the original SIPP, extended-grid, wall-time, and probe-reference runs evaluate sibling cohorts. All were re-executed on the frozen cohort under unchanged rules, identity verified by a serialized-world hash manifest (erratum in the appendix; the weighted-A* baseline always used the correct cohort).

Statistics and coverage. The smallest jointly solved cell is three maps; matched-effort magnitudes vary by pipeline on three suites; a best-configuration view in the appendix is post-hoc. Quoted intervals underrepresent training-run uncertainty (addressed dynamically via across-pipeline ranges); the discrete result covers selected cells; map-clustered reanalysis supports the quoted conclusions (not every exploratory table regenerated; no equivalence tests). The factorial covers one target family per domain, three adaptation seeds, and one sampled world set per cell (targeted redraws in the appendix); coverage counts worlds, not measured geometric diversity. Matching covers labels and steps, not world-generation cost; per-cell readouts are marginal (adjusted inference covers the 30 coverage contrasts; the two-world threshold is descriptive); the extended grid and probe/rescue are post-hoc sensitivity and diagnostic evidence. The primary dynamic experiments report expansions rather than end-to-end latency (timings from a 5-map-per-suite CPU sample; appendix).

Scope. The restricted-observation confirmation is static, one seed and cohort, tests input restriction only (the planner knows the roadmap), carries no formal bound on the direct arm, and is slower in wall time. Dynamic obstacles are deterministic; stochastic motion, kinodynamic constraints, multi-agent coupling, physical robots, and settings beyond 2-D navigation are untested. The external test coarsens and rectangle-decomposes five usable source maps, adds synthetic dynamics, and nests instance-level tests in those maps; SIPP is unbudgeted, its interval unit not resource-equivalent to (v, t) expansions. Negative results are failures to observe improvement under the tested protocols, not universal rejections; safety-critical use would need uncertainty-aware motion and collision guarantees.

Conclusion

A frozen blind U-Net transferred zero-shot on the procedural substrate: success improved on all six confirmation suites and matched effort fell 73–95%. Against success-tuned weighted A* the advantage is decisive on spiral, effort-favorable on four suites, and robust to the tested grid extension. Unbudgeted SIPP (implemented collision model) solves every feasible-labeled instance optimally, and the success advantage does not reproduce on the five MovingAI-derived maps (exploratory within-group adaptation restores dungeon success, street parity): the contribution is effort structure within budgeted search. Changing the supervised formulation or search-state coupling can reverse a planner-level result; low-rank updates protect at $K{=}1$, full fine-tuning overtakes on effort by $K{=}16$, and narrow world coverage drives the clearest held-out degradation in the tested factorial.

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